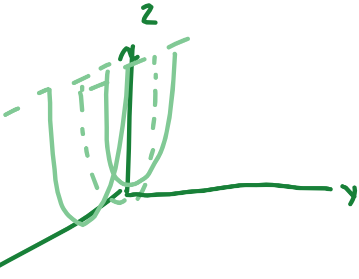
Surfaces in Space

Section 11.6

9-1-26

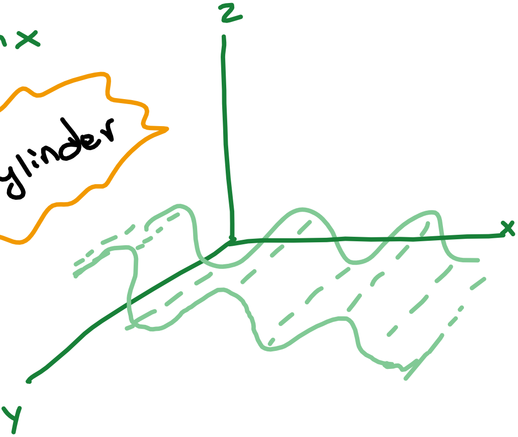
A) Sketching (by Trace)

ex) $z - y^2$



Parabolic Cylinder

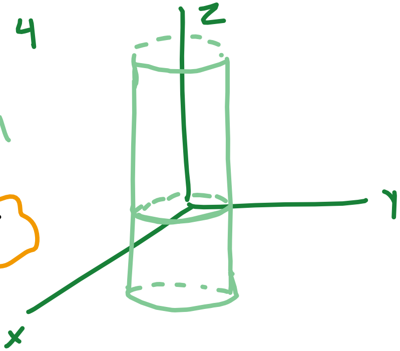
ex) $z = \sin x$



Sinusoidal Cylinder

ex) $4x^2 + y^2 = 4$

$$x^2 + \frac{y^2}{4} = 1$$



Elliptical cylinder

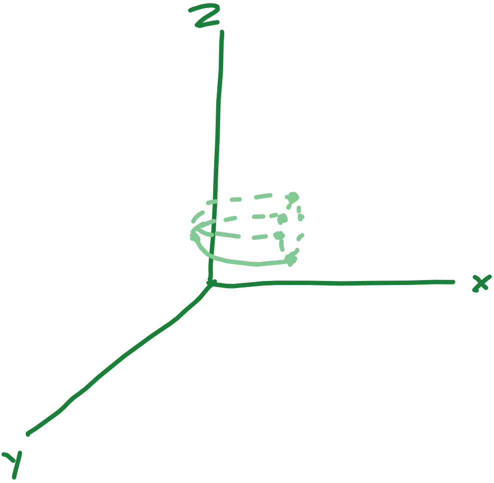
ex) $4y^2 + z^2 - x - 16y - 4z + 20 = 0$

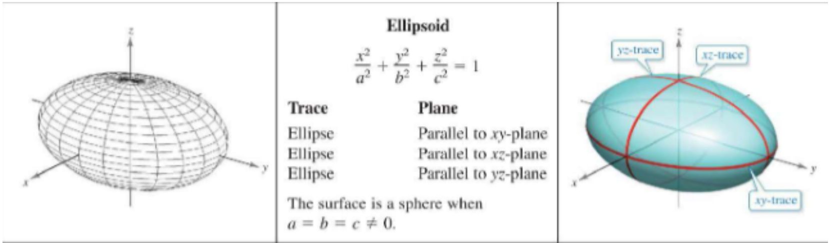
$4y^2 - 16y^2 + \_\_\_\_ + z^2 - 4z + \underline{4} = x - 20 + \underline{16} + \underline{4}$

$4\overset{\curvearrowright}{(y^2 - 4y + \underline{4})}$

$4(y-2)^2 + (z-2)^2 = x$

Parabaloid



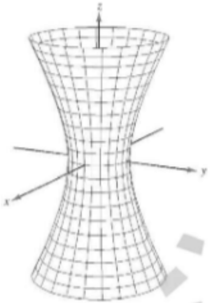

z
y
x
yz-trace
xz-trace
xy-trace

Ellipsoid

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

Trace	Plane
Ellipse	Parallel to xy-plane
Ellipse	Parallel to xz-plane
Ellipse	Parallel to yz-plane

The surface is a sphere when $a = b = c \neq 0$.

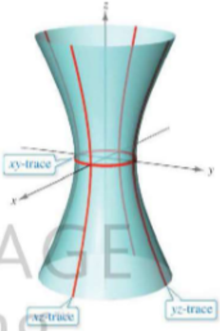


Hyperboloid of One Sheet

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}-\frac{z^2}{c^2}=1$$

Trace	**Plane**
Ellipse	Parallel to *xy*-plane
Hyperbola	Parallel to *xz*-plane
Hyperbola	Parallel to *yz*-plane

The axis of the hyperboloid corresponds to the variable whose coefficient is negative.


z
xy-trace
y
x
xz-trace
yz-trace

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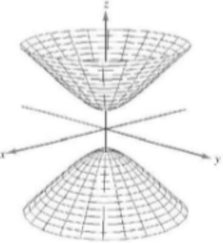
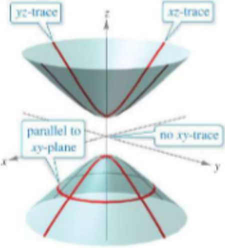


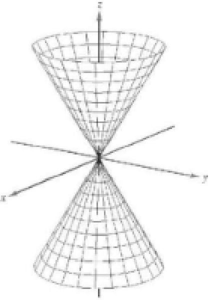
Hyperboloid of Two Sheets

$$\frac{z^2}{c^2}-\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$$

Trace	**Plane**
Ellipse	Parallel to *xy*-plane
Hyperbola	Parallel to *xz*-plane
Hyperbola	Parallel to *yz*-plane

The axis of the hyperboloid corresponds to the variable whose coefficient is positive. There is no trace in the coordinate plane perpendicular to this axis.


yz-trace
z
xz-trace
parallel to xy-plane
no xy-trace
x
y

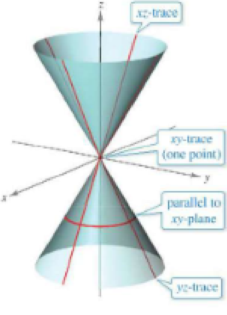


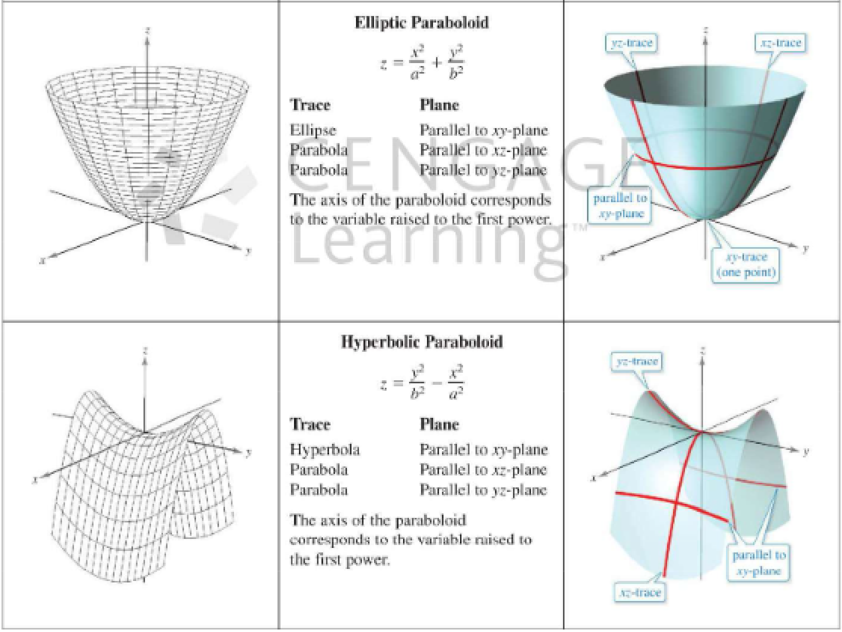
Elliptic Cone

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}-\frac{z^2}{c^2}=0$$

Trace	**Plane**
Ellipse	Parallel to *xy*-plane
Hyperbola	Parallel to *xz*-plane
Hyperbola	Parallel to *yz*-plane

The axis of the cone corresponds to the variable whose coefficient is negative. The traces in the coordinate planes parallel to this axis are intersecting lines.


z
xz-trace
xy-trace (one point)
y
x
parallel to xy-plane
yz-trace



Elliptic Paraboloid

$$z = \frac{x^2}{a^2} + \frac{y^2}{b^2}$$

Trace	Plane
Ellipse	Parallel to *xy*-plane
Parabola	Parallel to *xz*-plane
Parabola	Parallel to *yz*-plane

The axis of the paraboloid corresponds to the variable raised to the first power.

z
x
y
yz-trace
xz-trace
parallel to xy-plane
xy-trace (one point)

Hyperbolic Paraboloid

$$z = \frac{y^2}{b^2} - \frac{x^2}{a^2}$$

Trace	Plane
Hyperbola	Parallel to *xy*-plane
Parabola	Parallel to *xz*-plane
Parabola	Parallel to *yz*-plane

The axis of the paraboloid corresponds to the variable raised to the first power.

z
x
y
yz-trace
parallel to xy-plane
xz-trace

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11.6 HW 9/2

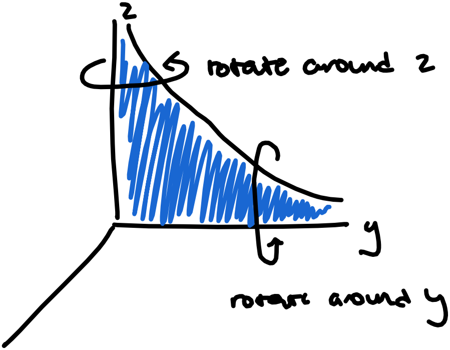
1) c 2) e 3) f 4) b 5) d 6) a

21) VII 22) IV 23) II 24) III

25) VI 26) I 27) VIII 28) V

B) Surface of Revolution

1. about x-axis $y^2 + z^2 = f(x)^2$
2. about y-axis $x^2 + z^2 = f(y)^2$
3. about z-axis $x^2 + y^2 = f(z)^2$


z
rotate around z
y
rotate around y

---

ex) find the equation for surface of revolution of $y = \frac{1}{z}$ about the z-axis

$$x^2+y^2=\big(\underset{\uparrow\ \text{generating curve}}{f(z)}\big)^2$$

$$x^2+y^2=\left(\frac{1}{z}\right)^2$$

or

$$x^2z^2+y^2z^2=1$$

---

ex) graph $9x^2=y^3$

about y-axis:

$$x^2+z^2=(f(y))^2 \qquad x^2+z^2=\frac{y^3}{9}$$

about x-axis:

$$y^2+z^2=(f(x))^2 \qquad y^2+z^2=(9x^2)^{2/3}$$

---

ex) graph $x=e^{-y}$ | ex) graph $z=e^{-y}$

about y-axis

$x^2+z^2=(f(y))^2$

$x^2+z^2=(e^{-y})^2$

about y-axis

$x^2+z^2=(f(y))^2$

$x^2+z^2=(e^{-y})^2$

The Same!

c) Find a generating curve

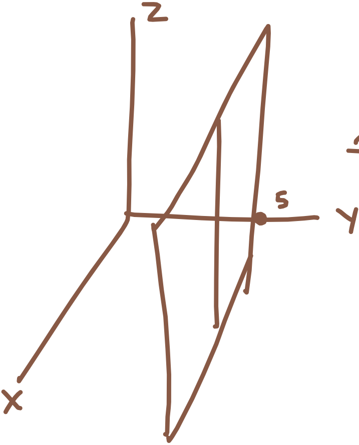
ex) $x^2+y^2+z^2=9$

$x^2+z^2=9-3y^2$

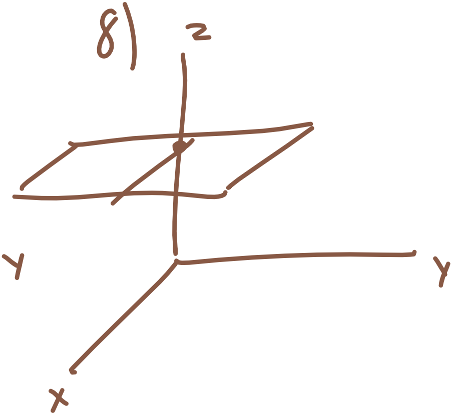
$x=\sqrt{9-3y^2}$ $z=\sqrt{9-3y^2}$

11.6 HW 9/3

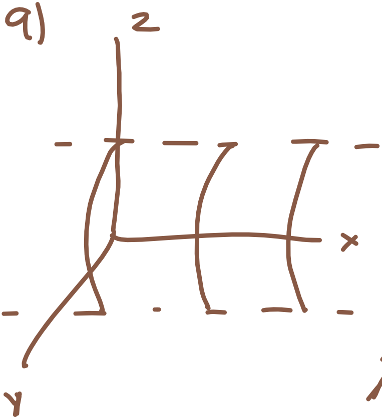
7)



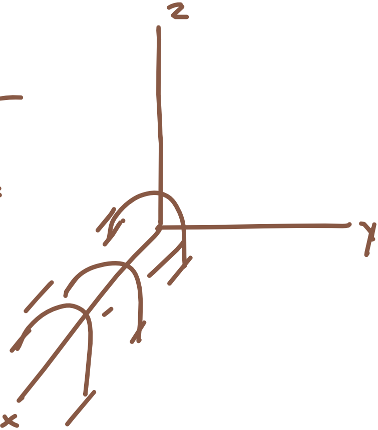
8)



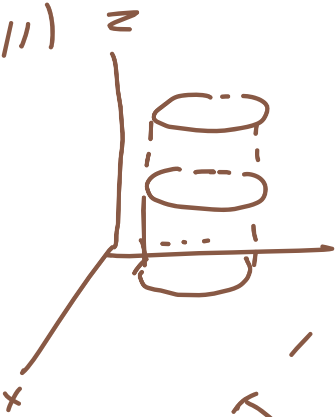
9)



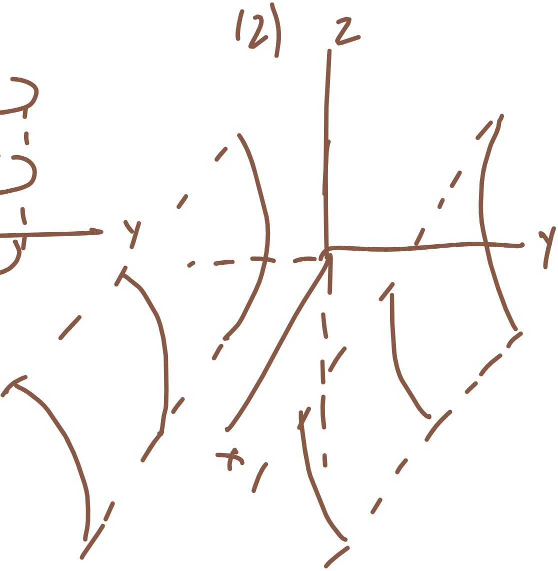
10)



11)



12)



31) $z^2 = 4y$ about y-axis

$$x^2 + z^2 = (f(y))^2$$

$$\boxed{x^2 + z^2 = 4y}$$

33) $z = 2y$ about z-axis

$$x^2 + y^2 = (f(z))^2$$

$$\boxed{x^2+y^2=\frac{z^2}{4}}$$

35) $xy=2$ about x-axis

$$y^2+z^2=(f(x))^2$$

$$\boxed{y^2+z^2=\left(\frac{2}{x}\right)^2=\frac{4}{x^2}}$$

38) $x^2+z^2=\cos^2 y$

$$\boxed{x=\cos y \text{ or } z=\cos y}$$

48) $2(x^2+3y^2-2z^2+2y=4)$

$$2x^2+6y^2-4z^2-3x=2$$

$4y+3x=6$

this creates a plane in only y and x in 3d space

49) $\boxed{\text{True}}$

50) $\boxed{\text{False}}$

$$x^2+y^2=\cos^2 z$$

$y=\cos z$ $\quad$ $x=\cos z$

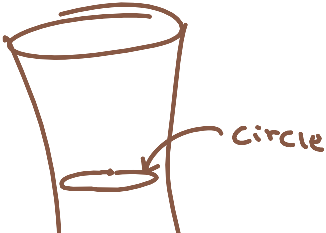
$$2x^2+2y^2=2\cos^2 z$$

$z=\cos y$ $\quad$ $x=\cos y$

51) $\boxed{\text{False}}$

trace $x=2$ of $\frac{x^2}{4}+\frac{y^2}{9}+z^2=1$

is $(2,0,0)$

52) $\boxed{\text{False}}$